

DATA ANALYTICS INTERNSHIP PROJECT

Retail Sales Analytics and Business Performance Optimisation

SuperStore Dataset | Python | Power BI

Submitted to

MashupStack — Data Analytics Training Programme

PROJECT TEAM

Member	Tasks Handled
Harigovind	Task 1, Task 4, Task 6
Amritha	Task 2 — EDA
Devika	Task 3 — RFM Analysis
Keerthana	Task 5 — Visualisation & Dashboard

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Executive Summary

This report presents a comprehensive data analytics project conducted on the SuperStore retail dataset. The project was carried out as part of the MashupStack Data Analytics Internship Programme. A team of four members worked collaboratively across six structured tasks, covering data cleaning, exploratory analysis, customer segmentation, business investigation, dashboard development and final strategic recommendations.

The dataset contains 51,290 order records spanning multiple regions, product categories and customer segments. Analysis was performed using Python (pandas, matplotlib, seaborn) and Power BI. Key findings include a critical discount policy issue — orders with discounts above 20% consistently generate losses — and significant regional underperformance in Southeast Asia and EMEA. The report concludes with six areas of actionable business recommendations.

Metric	Value
Total Orders Analysed	51,290
Total Sales Revenue	12,642,905
Total Profit	1,468,905
Overall Profit Margin	11.62%
Regions Analysed	13
Product Categories	3
Customer Segments (RFM)	4
Loss-Making Sub-Category	Tables (-8.47% margin)
Most Profitable Region	Canada (26.62%)
Critical Discount Threshold	20% — above this, avg loss per order

Task 1**Data Understanding and Cleaning**

Harigovind

Overview

The raw SuperStore dataset was loaded and inspected for quality issues before any analysis was performed. Three key data problems were identified and resolved and five new columns were engineered to support downstream analysis.

Issues Found and Fixed**Issue 1 — Sales Column: Wrong Data Type**

The sales column was stored as a string with comma-formatted numbers (e.g., '1,234.56'). This prevented any arithmetic operations. Commas were stripped and the column was converted to a numeric (integer) type using pandas.

Fix applied: `pd.to_numeric(df['sales'].str.replace(',', ''), errors='coerce')`

Issue 2 — Date Columns: Stored as Strings

Both `order_date` and `ship_date` were stored as plain strings rather than datetime objects, making time-based analysis impossible. Both columns were parsed using mixed-format datetime conversion.

Fix applied: `pd.to_datetime(df['order_date'], dayfirst=True, format='mixed')`

Issue 3 — Year Column: Inconsistent with Order Date

The existing year column contained 31,223 records that did not match the year derived from `order_date`. The column was rebuilt from scratch using the corrected datetime column.

Fix applied: `df['year'] = df['order_date'].dt.year`

New Columns Created

Column	Description	How Derived
Profit Percentage	Profit as % of sales	$(\text{profit} / \text{sales}) * 100$
Order Month	Month extracted from <code>order_date</code>	<code>order_date.dt.month</code>
Order Year	Year extracted from <code>order_date</code>	<code>order_date.dt.year</code>
Discount Level	Categorical discount band	None / Low / Medium / High
Customer Lifetime Value	Total spend per customer	Grouped sum of sales by customer

Tools Used

- Python 3
- pandas

Task 2**Exploratory Data Analysis (EDA)**

Amritha

Overview

Exploratory Data Analysis was performed across four dimensions: sales and profit, time-based trends, product performance and customer behaviour. All visualisations were created using matplotlib and seaborn in Jupyter Notebook.

A — Sales and Profit Analysis

Sales and profit were broken down by category, sub-category, region, state and customer segment. Technology was the highest-revenue category, while Office Supplies and Technology both achieved profit margins close to 14%. Furniture lagged significantly at 6.98%.

Category	Total Sales	Total Profit	Profit Margin
Furniture	4,110,884	286,782	6.98%
Office Supplies	3,787,330	518,474	13.69%
Technology	4,744,691	663,779	13.99%

B — Time-Based Analysis

Monthly profit analysis revealed a clear seasonal pattern. The second half of the year — particularly September through December — accounts for the majority of annual profit. November is consistently the strongest month.

Period	Months	Profit Share	Strategy Note
Peak Season	Sep – Dec	~45% of annual profit	Pre-stock high-margin products
Mid Season	May – Aug	~33% of annual profit	Push targeted campaigns
Low Season	Jan – Apr	~22% of annual profit	Focus on Paper, Labels, low-cost lines

C — Product Performance

The top 10 profitable products were dominated by Copiers, high-end Technology accessories and premium Binders. The bottom 10 loss-making products were concentrated in the Tables and Machines sub-categories, where heavy discounting eroded all margin.

D — Customer Analysis

Top customers by revenue were identified and cross-referenced with profitability. Several high-revenue customers were found to be generating negative profit — a key finding investigated further in Task 4. Repeat purchasing behaviour showed that a small group of customers drives a disproportionate share of revenue.

Tools Used

- pandas
- matplotlib
- seaborn
- Jupyter Notebook

Task 3

Customer Segmentation — RFM Analysis

Devika

Overview

RFM (Recency, Frequency, Monetary) analysis was applied to segment the customer base into four distinct groups. Each component was scored from 1 to 4 and combined into a composite RFM score to classify customers.

RFM Components

Component	Definition	Business Meaning
Recency (R)	Days since last purchase	Lower = more engaged recently
Frequency (F)	Number of orders placed	Higher = more loyal
Monetary Value (M)	Total amount spent	Higher = more valuable

Customer Segments

Segment	Count	Characteristics	Priority
High-Value Customers	189	Recent, frequent, high spend	Highest
Regular Customers	339	Consistent buyers, moderate spend	High
At-Risk Customers	175	Previously active, now lapsing	Medium
Low-Engagement Customers	92	Infrequent visits, low spend	Low

Business Interpretation

High-Value (189): These customers are the backbone of revenue. They purchase frequently and recently. Losing even 10% of this segment would have a significant impact on total profit.

Regular (339): The largest group. Consistent buyers who can be moved to High-Value through cross-selling, upselling and loyalty programmes.

At-Risk (175): Previously engaged customers who have not purchased recently. Win-back campaigns and personalised outreach should be prioritised for this group.

Low-Engagement (92): Infrequent, low-spend customers. Marketing spend on this group should be minimal — low-cost email re-engagement only.

Task 4

Business Problem Investigation

Harigovind

Overview

Five real business problems were investigated using data. Each finding is supported by quantitative evidence and followed by a specific recommendation.

Q1 — Why Do Some Categories Have High Sales But Low Profit?

Furniture generates the second-highest sales volume but achieves a profit margin of only 6.98%, roughly half the margin of Technology and Office Supplies. At the sub-category level, Tables is actively loss-making at -8.47% margin — the company loses money on every table sold.

Sub-Category	Sales	Profit	Margin	Status
Tables	757,034	-64,083	-8.47%	LOSS
Machines	779,071	58,868	7.56%	LOW
Chairs	1,501,682	141,974	9.45%	LOW
Copiers	1,509,439	258,568	17.13%	GOOD
Paper	244,307	59,208	24.23%	BEST

INSIGHT: The Tables sub-category alone generated a total loss of 64,083. This is caused by excessive discounting and high shipping costs relative to the product's selling price.

RECOMMENDATION: Review Tables pricing. Apply a minimum margin rule or discontinue the product line. Shift investment to Paper (24.23%) and Copiers (17.13%).

Q2 — Are Discounts Negatively Impacting Profit?

Discount levels were segmented into four bands and compared against average profit per order. The discount-profit correlation is -0.32, confirming a clear negative relationship.

Discount Level	Range	Average Profit	Verdict
None	0%	+61.04	Profitable
Low	1-20%	+46.84	Profitable
Medium	21-50%	-56.19	LOSS
High	51-85%	-98.89	HEAVY LOSS

INSIGHT: Any order with a discount above 20% produces an average loss. Medium discounts generate -56 per order; high discounts cost the company -99 per order.

RECOMMENDATION: Cap all discounts at 20% company-wide. Require manager approval for any exception.

Q3 — Which Regions Require Business Intervention?

Profit margin was calculated across all 13 regions. Southeast Asia and EMEA emerged as the most critical cases.

Region	Sales	Profit	Margin	Status
Southeast Asia	884,438	17,852	2.02%	URGENT
EMEA	806,184	43,898	5.45%	URGENT
South	1,600,960	140,356	8.77%	IMPROVE
Central	2,822,399	311,404	11.03%	MONITOR
North	1,248,192	194,598	15.59%	HEALTHY
Canada	66,932	17,817	26.62%	BEST

INSIGHT: Southeast Asia generates 884,438 in sales but only 17,852 in profit — a 2.02% margin. EMEA is similarly poor at 5.45%.

RECOMMENDATION: Conduct an immediate cost and discount audit in Southeast Asia and EMEA. Restrict discount approvals in these regions to management level only.

Q4 — Which Products Should Receive More Marketing Focus?

Products were classified into four quadrants based on sales volume and profit relative to the dataset median.

Quadrant	Count	Definition	Action
Star Products	1,372	High sales + High profit	Maintain and promote
Hidden Gems	522	Low sales + High profit	Increase marketing
Review Pricing	524	High sales + Low profit	Fix discount or cost
Consider Discontinue	1,370	Low sales + Low profit	Evaluate exit

INSIGHT: 522 products have strong profit margins but low sales. These hidden gems represent the easiest profitability gain — they just need visibility.

RECOMMENDATION: Allocate 30% of marketing budget to Hidden Gem products. Feature them in email campaigns, homepage banners and bundles with Star products.

Q5 — Are Customers Generating High Sales But Low Profit?

23 customers were identified who generate above-median sales revenue but produce negative profit. These customers effectively cost the company money despite appearing as high-value accounts.

Customer	Total Sales	Total Profit	Margin
Grant Thornton	20,226	-3,578	-17.69%
Denise Monton	22,051	-2,598	-11.78%
Sean Braxton	20,514	-1,897	-9.25%
Michelle Tran	17,779	-1,788	-10.06%
Henry Goldwyn	17,168	-1,598	-9.31%

INSIGHT: These customers receive excessive personalised discounts that push every order into negative territory. High revenue does not equal profitable revenue.

RECOMMENDATION: Remove special discount privileges from all 23 problem customers. Apply a minimum 5% profit margin rule per customer account.

Task 5

Visualisation and Dashboard Development

Keerthana

Python Visualisations

Meaningful charts were created using matplotlib and seaborn across all analysis tasks. Each visualisation was tied to a specific business question.

Chart	Type	Business Question Answered
Category Sales vs Profit vs Margin	Grouped Bar	Why does Furniture underperform?
Sub-Category Profit Margin	Horizontal Bar	Which sub-categories are loss-making?
Discount vs Profit	Scatter	Do discounts hurt profit?
Avg Profit by Discount Level	Bar	At what discount level do losses begin?
Regional Profit Margin	Bar	Which regions need intervention?
Region vs Category Heatmap	Heatmap	Where are the worst margin combinations?
Product Quadrant Classification	Scatter	Which products need marketing?
Customer Sales vs Profit Classification	Scatter	Which customers are loss-generating?
Monthly Profit Trend	Bar	What is the seasonal profit pattern?
RFM Customer Segment Distribution	Bar + Pie	How is the customer base distributed?

Power BI Dashboard

An interactive Power BI dashboard was developed containing the following components:

- Total Revenue KPI card
- Total Profit KPI card
- Profit Margin KPI card
- Regional performance analysis — map and bar visuals
- Product and category performance breakdown
- Customer segmentation insights from RFM output
- Monthly sales trend — line chart
- Discount vs Profit analysis — scatter with trend line

The dashboard includes filters and slicers for Region, Category, Segment and Year, allowing management to drill into any combination of dimensions interactively.

Task 6

Final Business Recommendations

Harigovind

Overview

Based on all findings from Tasks 1 through 5, six areas of actionable recommendations are presented below. Each recommendation is directly supported by data.

1 — What the Company Should Stop Doing

- Stop offering discounts above 20%. Medium and High discount levels consistently produce losses — average -56 and -99 per order respectively.
- Stop blanket discount campaigns. Discounts should be targeted and margin-checked before approval.
- Stop selling Tables at current pricing. The sub-category has a -8.47% margin and generated a total loss of 64,083.
- Stop ignoring Southeast Asia and EMEA. Both regions are critical underperformers at 2.02% and 5.45% margins.
- Stop granting special discount privileges to the 23 identified loss-generating customer accounts.

2 — Products and Categories to Promote

The following sub-categories have profit margins above 15% and should receive increased marketing investment:

Sub-Category	Profit Margin	Recommended Action
Paper	24.23%	Feature in campaigns, bundle with printers
Labels	20.44%	Cross-sell with Office Supplies orders
Envelopes	17.32%	Include in B2B and stationery kits
Accessories	17.30%	Bundle with Technology products
Copiers	17.13%	Key focus for corporate and B2B segment
Binders	15.68%	Bundle with Paper and Art sub-categories

3 — Regional Improvement Strategies

URGENT — Southeast Asia (2.02%): Immediate discount and cost audit. Restrict approvals to management only.

URGENT — EMEA (5.45%): Review supplier contracts and operational overhead. Identify loss-driving sub-categories in this region.

IMPROVE — South (8.77%): Shift product mix toward higher-margin sub-categories. Reduce Tables and Machines volume.

EXPAND — Canada (26.62%) and North Asia (19.52%): These are the healthiest regions. Study their pricing discipline and product mix and replicate across underperforming regions.

4 — Customer Segments to Target

Segment	Count	Strategy
High-Value	189	Loyalty programmes, early product access, dedicated support — do not offer heavy discounts
Regular	339	Cross-sell and upsell campaigns, bundle deals, controlled 10-15% loyalty discounts
At-Risk	175	Win-back campaigns with personalised offers and feedback surveys
Low-Engagement	92	Low-cost email re-engagement only — avoid heavy marketing spend

5 — Pricing and Discount Strategy

Category / Rule	Current Practice	Recommended Cap
Technology	Up to 85%	Maximum 15%
Office Supplies	Up to 85%	Maximum 15%
Furniture	Up to 85%	Maximum 10%
Tables (sub-cat.)	Up to 85%	0% — no discount until profitable
Any discount >20%	No approval	Manager sign-off required
Per customer	Ad hoc	Minimum 5% margin per account enforced

6 — Suggestions to Improve Profitability

- Seasonal planning: Pre-stock high-margin products before the September–December peak. Focus on low-cost, high-margin items (Paper, Labels) in January–April.
- Supplier renegotiation: Furniture's 6.98% margin suggests supplier costs are too high. Renegotiate contracts or source alternative suppliers.
- Shipping optimisation: Route low-value orders through Standard shipping. Restrict First Class and Same Day to high-value or time-sensitive orders only.
- Hidden gem marketing: The 522 hidden gem products with high margins and low sales represent the easiest profitability gain. Allocate 30% of marketing budget here.
- Referral programme: Launch a referral programme for High-Value customers to attract similar-profile buyers.
- Bundling strategy: Bundle Technology products with Office Supplies to increase average basket value without discounting either item.

Final Deliverables

#	Deliverable	Format	Status
1	Cleaned Dataset	.csv / .xlsx	Completed
2	Python Notebooks	.ipynb	Completed
3	Power BI Dashboard	.pbix	Completed
4	Final Project Report	.docx / .pdf	Completed

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